Common Logical Errors

In C Programming Language

1. Condition Error
2. Iteration Error
3. Boundary Error
4. Formula Error
5. Initialisation Error
6. Datatype Error
7. Invalid Input Error

Logical Error: *A logical error is a flaw in the program logic that does not prevent execution but produces incorrect output.*

Condition Error

Wrong condition used in decision making.

Example: Check if number is 50 or above

Faulty
```
if (marks > 50)
// Wrong relational operator used
```

Robust
```
if (marks >= 50)
// This will check if marks equal 50
```

Iteration Error

Incorrect loop logic or update.

Example: Print numbers from 1 to 5

Faulty
```
i=1;
while (i <=5)
    printf("%d", i);
// Loop control variable not updated
```

Robust
```
i=1;
{
while (i <=5)
    printf("%d", i);
    i++;
}
//Increment exits loop
```

Boundary Error

Off-by-one error in loop boundary

Example: Access array of size 5

Faulty
```
for (i = 0; i <= 5; i++)
// Array index out of bounds
```

Robust
```
for (i = 0; i < 5; i++)
// Including 0, the size of the array is 5
```

Formula Error

Incorrect formula usage

Example: Average of 2 variables

Faulty
```
avg = a + b / 2;
// This does not follow PEMDAS/BODMAS
```

Robust
```
avg = (a + b) / 2;
// Yields the average of variables a and b
```

Initialisation Error

Uninitialized declared variable holds garbage value

Example: Sum of a and b

Faulty
```
int a, b = 5;
int sum = a + b;
//Addition of 5 to a garbage value
```

Robust
```
int a = 10, b = 5;
int sum = a + b;
//Addition of desired 2 integers
```

Datatype Error

Implicit type conversion may cause precision loss

Example: Add decimal numbers

Faulty
```
int sum;
sum = 1.2 + 3.4;
//Sum stores result 4 (Discards the decimal)
```

Robust
```
float sum;
sum = 1.2 + 3.4;
//Sum stores result 4.6
```

Invalid Input Error

Lack of input constraints leads to logical issues

Example: Check if user is minor

Faulty
```
int age;
scanf("%d", &age);
if (age < 18)
// This would allow age to be negative
```

Robust
```
int age;
scanf("%d", &age);
if (age > 0 && age < 18)
// This infers integer ages from 1 to 17
```


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